

L4.1b Graphing Functions of y

Sideways graphing — what you need to know

The swap

The input runs **UP** the page. The output runs **ACROSS** it. Every row below is a consequence of that one line.

You know this for $y = f(x)$

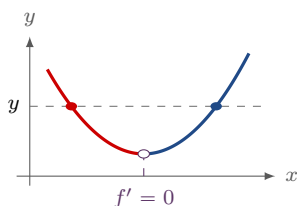
slope = m : rise over run
 quadratic, $a > 0$ opens up; $a < 0$ opens down
 the zeros are x -intercepts
 $f(0)$ is the y -intercept
 Loh's midline m : the vertical line $x = m$
 denominator = 0: vertical asymptote
 limit at $\pm\infty$: horizontal asymptote
 vertical line test

Same idea for $x = g(y)$

$m = \Delta x / \Delta y$: **run over rise**
 opens **right**; opens **left**
 the zeros are y -intercepts
 $g(0)$ is the x -intercept
 the **horizontal** line $y = m$; axis of symmetry horizontal
horizontal asymptote
vertical asymptote
horizontal line test

Transformations are unchanged. *Outside affects the output normally; inside affects the input backward.*
 Outside now moves x , across. Inside still runs backward.

MONOTONIC PIECES



$x = f^{-1}(y)$ exists on any interval where f is monotonic

“every horizontal line crosses once, so each y comes from one x ”

Not monotonic on the interval? **Split at $f'(x) = 0$.** On each piece it is, and each piece solves for x — the $\pm$ branches are the pieces.

WATCH OUT

✗ $g(0)$ is the y -intercept $\rightarrow$ the x -intercept — input 0 is the horizontal axis, and the output lands on it

✗ the slope of $x = 2y + 1$ is really $\frac{1}{2} \rightarrow$ read it as $\Delta x / \Delta y = 2$: up 1, across 2

✗ turn the page to make the graph upright $\rightarrow$ nothing was reflected — you changed which variable you chose

✗ $x = 1 \pm \sqrt{1 + y}$ is one function of $y \rightarrow$ two — one branch per monotonic piece, split where $f'(x) = 0$